

When Should rare Win—and Lose?

Conditional directions only. No numeric result, method rank, or gate closure is asserted.

This v5 guide repairs the link between the proposed mechanism and the paper-facing experiment tables. It keeps the v4 benchmark–baseline closure contract and adds an explicit regime-by-contrast direction matrix. A direction is eligible for confirmation only after its premise, read cut, common evaluation population, cost envelope and falsifier are frozen on a development stream or by an exogenous mutation. Conditions must never be named after seeing the confirmation result.

Status and interpretation

The current status is `CURRENT_CONDITIONAL_DIRECTION_GUIDE_PENDING_EVIDENCE`. The current evidence has no qualified confirmation roots, complete live baseline parity, independent future assignment outcomes, or scientific closure receipt. The v1 numerical targets were withdrawn and cannot be reused. The synthetic receipt tests validate only the software contract, not this matrix or a method effect.

Symbol	Meaning	Rule for use
>	RARE should be better than the named baseline on this endpoint	Only after the premise and parity contract are sealed.
=	Pre-registered equivalence or non-inferiority relation	A margin and paired estimator are required; non-significance is not equivalence.
<	RARE should be worse in this negative regime	A deliberate loss boundary, retained as evidence.
?	Two-sided or not identifiable before data	No superiority claim; estimate both signs.
N/A	Arm or endpoint is not qualified	Preserve a NO-GO reason; an empty cell is not a ranking.

Four independent endpoints

H1 is Brier/log loss on the same already-selected held-out cohort (lower is better), with producer contract Q_p and later-use Y kept separate. H2 is independent future-assignment utility $U = Y - \lambda C$ (higher is better), with all-attempt ITT denominators and UNKNOWN bounds. H3 is full publish/read/update p95, backlog and drop/timeout rate under a fixed arrival and resource envelope (lower is better). H4a is false producer attribution with independent owner truth (lower is better). Coverage, abstention, UNKNOWN and owner-truth recall are H4b companion quantities; abstention cannot be used to manufacture an H4a win.

Table 1: Baseline and Factor Qualification

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Every row below is a required arm or a deliberately retained NO-GO. The same candidate menu/version, legal read cut, public representation, encoder/checkpoint/schema, opportunity set, delay, exploration, capacity, service/API/token/tool/state/queue budget and complete cost must be used. Each adaptive arm keeps its own independent realized-history namespace; a pooled controller pools only inside its own policy.

Arm	Role	Required qualification	If absent
Uniform / no update	Primary weak control	Same material-generation path and opportunity schedule; no learned state	RARE cannot claim learning over a prior.
Raw acceptance / terminal-only	Signal controls	Same public trace and target cohort; isolate raw verdict and terminal evidence	H1/H2 information increment unidentified.
Contextual-trust-linear	Strong same-info control	Full public ϕ , same capacity/read cut/cost/history opportunity and own state	No ordinary contextual-learning claim.
Task-conditioned ridge	Strong matching control	Task-peer interactions, same gate/trace/menu and live parity; identity-only ridge is insufficient	Task-peer gain is unidentifiable; retain NO_GO.
Pooled controller	Data-pooling control	Own namespace and predeclared capacity; cannot read another arm's live state	Local personalization claim is confounded.
RARE frozen card	Candidate	Exact representation, head/updater, gates, delay and service contract sealed	No method-level latency/forgetting claim.
Closest published adapter	Timeliness/reference	Faithful same-protocol adapter, or explicit NO_GO with reason	Published comparison cannot support superiority.
Count-only / J-masked / trace-only / J-permutation	Information counterfactuals	Same trace/menu/version/read cut; selected cohort and costs retained	J-specific claim fails closed.

Mandatory factorial and locality blocks

Evidence and delayed credit are one four-arm matched factorial: (i) neither, (ii) public evidence only, (iii) delayed credit only, (iv) both. The interaction $I_U = U_{11} - U_{10} - U_{01} + U_{00}$ uses the qualified stream covariance. Locality requires local, nonlocal/global, and same-cost centralized diagnostics. Graph-IPD is an external native cooperation reference and cannot fill these cells without a qualified adapter. Updater comparisons (RLS, online SGD/logistic, periodic refit) are nested under the same information policy; they are not silently omitted from a claim about update speed or recovery.

Table 2: Conditional Direction Matrix (H1/H2)

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Cells are the relation of RARE to the named baseline. The premise column is part of the experiment card. A condition not established before confirmation is ‘?’ for all affected contrasts, regardless of the observed result.

Regime	Contrast	H1	H2	H3	H4a	Falsifier / allowed interpretation
Typed informative evidence	Uniform / no update	>	>	?	>	J and typed ownership/version fields add signal beyond controls; trace/J masking or matched composition reproducing the gain falsifies mechanism superiority.
Typed informative evidence	Contextual trust / task ridge	>	>	?	>	Only after the strong same-info control is qualified; otherwise this is not an identified gain.
Trace explains defect	Trace-only	=	=	<	?	C1-like public execution facts already explain the defect; any gain after J masking suggests leakage or unequal inputs.
Stable task-peer crossover	Pooled controller	>	>	?	?	Requires repeatable crossover and sufficient local support on root/time holdout; pooling may win when data are sparse.
Constant J / cold start	Uniform / no update	=	<	<	=	Extra evidence/update cost has no signal to buy; a surviving J-specific gain is a hard leakage check.
Stale / version mismatch	Version-aware reset	<	<	?	>	Version isolation may reduce false credit while slowing recovery; a control missing the version field is invalid.
Sparse / shifted semantics	Pooled or task ridge	<	<	?	?	Personalization has high variance and negative transfer; repeated-template-only gains fail the holdout.
High arrival / evidence cost	Matched local uniform	?	<	<	?	Quality may rise while queue/backlog and complete cost erase utility; core update time is insufficient.
Gradual legal drift	Frozen prior	>	>	?	?	Drift schedule and delayed labels are sealed; report recovery separately from old-root forgetting.

The first two rows are the only candidate-positive regimes. They require a development test that typed responsibility/version information is incrementally predictive. Against the strong contextual/task-conditioned control, ‘>’ is not assumed merely because the candidate uses a learned ledger. A tie or a control win is an explicit method falsifier.

Table 3: Conditional Direction Matrix (H3/H4 and Stability)

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The method is a protocol with a replaceable implementation until the frozen card names the encoder, head, updater, capacity and service path. Therefore H3 and continual learning directions are not inferred from a 0.297 ms core update or from a native Graph-IPD score. They require end-to-end receipts.

Regime	Contrast	H3 p95	H3 backlog	H4a	Stability interpretation
Informative, sufficient support	Strong same-info control	?	?	>	Quality evidence may cost more; report a Pareto point, not a universal speed win.
Trace-only explanation	Trace-only	<	<	?	Extra J/retrieval is a service loss unless it buys independent future utility.
Constant J / cold start	No update	<	<	=	Same attribution semantics with extra work; no safety gain is presumed.
Stale replacement	Version-aware reset	?	?	>	Safety gate may abstain and lower coverage; report coverage and recovery cost together.
Sparse local history	Pooled / ridge	?	?	?	State or communication cost can dominate; no direction is sealed without the implementation card.
High arrival / queue pressure	Matched local uniform	<	<	?	Drops/timeouts and failed attempts stay in the denominator and cost ledger.
Gradual drift	Matched RLS/SGD/refit	?	?	?	Separate old-root forgetting from recovery delay; updater superiority is not automatic.

Stability and safety definitions

Old-root forgetting is the drop from the pre-drift held-out performance on the old root. Recovery delay is time/steps to regain a predeclared threshold on the new root. They are separate endpoints. State bytes is a resource axis, not a success direction: lineage-safe provenance can require more state. False attribution, false negatives, coverage, abstention and UNKNOWN must be reported jointly, stratified by producer-only, recipient-only, sink-only, mixed ownership and resource failure.

Table 4: Result Entry and Closure

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No direction row can be entered as a result until the root, stream, arm, factor and future outcome fields below are present. This is the ledger that connects a compact paper table to a real receipt.

Field	Required evidence	Failure interpretation
Root / stream	At least three structural roots for proceedings-ready, or a pre-registered precision-justified two-root design; independent realized histories and structural/time holdout	Seeds or topology labels alone do not create roots.
Common population	Same held-out cohort for H1, or randomized transport with overlap/weight uncertainty; Q_p and Y separate	Policy-dependent selected populations make Brier/log loss non-identifiable.
Future Y	Sealed assignment before target execution and blinded independent scorer; missing/partial is UNKNOWN	No future Y means H2 is UNSTUDIED, never neutral.
Denominators	ITT all attempts, started/completed/known/unknown/failed/unstarted, selected/unselected and coverage by owner stratum	Dropping UNKNOWN or hard cases can reverse ranking.
Cost	Selection, publish, read, judgment, scorer, update, assignment, communication, retry, tool/API/token, state, queue, repair and replay costs plus wall time	Quality gains that lose after complete cost are trade-offs.
Statistics	Paired stream/root effects, covariance for U, p95 tail estimator, UNKNOWN worst/best bounds, multiplicity/stopping and predeclared MDE/margin	A single aggregate or normal error bar is insufficient.
Falsifiers	J permutation, trace-only, menu/id permutation, version swap, recipient-only/mixed/UNKNOWN, unselected-label censoring, replay idempotence and gold future-label injection	Any hard-fail invalidates the closure attempt; preserve raw evidence.
Replay/review	Independent environment/code path, blind scorer, hashes, raw event chain, clean replay and independent scientific reviewer	Software validator pass alone cannot close the scientific gate.

Closure rule

The v4 B1–B7 receipt remains the sole benchmark+baseline closure rule. Guide alignment is necessary and not sufficient. The scientific gate is **OPEN** until all required roots, live baseline parity, independent streams, future-Y cells, complete tables, statistics/cost, clean replay and independent review are present and verified. A full positive, neutral or adverse result can close the evidence gate; an incomplete, **UNSTUDIED** or findings-ready result cannot. A closed evidence gate never turns a null or adverse method result into a positive claim.